Zikai Zhao

+86-18833116940 | zikai3@illinois.edu
+1-217-904-2935 | zikai1.23@intl.zju.edu.cn
Personal website: webzikaizhao.github.io

Education

Zhejiang University — Zhejiang, China
B.Eng., ***GPA: 4.17/4.3*** — *Sep 2023 – Jun 2027*
Major: Electrical Engineering and Automation (Dual Degree)

University of Illinois Urbana-Champaign — Urbana, IL, USA
B.S., ***GPA: 3.96/4.0*** — *Sep 2023 – Jun 2027*
Electrical Engineering (Dual Degree)
Minor in Computer Science (requirements completed)

Research Experience

Physics-Informed Learning for Drone Dynamics — Jun 2026 – Present
Supervisor: Prof. Fadi Kurdahi, UCI
- Develop and train dynamics models on the NeuroBEM dataset for long-horizon trajectory prediction, incorporating symmetry, monotonicity, zero-state constraints, and trajectory-error penalties.
- Conduct ablation studies to evaluate individual physical constraints and their integration into training; identify mismatched symmetry assumptions and test-coverage limitations affecting their benefit.
- On challenging test cases, reduced tail-segment trajectory error by approximately one-third versus a one-step, data-only baseline and to one-quarter of the error of a physics-only baseline.

Fault Diagnosis in Kolmogorov–Arnold Networks (KANs) — Jun 2026 – Present
Supervisor: Prof. Fadi Kurdahi, UCI
- Investigate fault localization through single-parameter error injection, carrying out experiments with regular feedback from senior researchers in a three-person team.
- Switched from classification to function-fitting tasks to reduce baseline loss and make fault-induced error regions easier to detect.
- Observed input-localized error from single-parameter faults in function-fitting experiments, narrowing candidate nodes or layers; compared with MLP baselines and identified KAN runtime limitations.

4-Bit Absolute Value Detector Design and Optimization — Sep 2024 – Dec 2024
Supervisor: Prof. Dejan Markovic, UCLA
- Independently derived a minimal Boolean design from a 128-entry truth table and optimized gate topology and sizing, achieving a minimum delay of approximately 33.7 delay units.
- Jointly optimized supply voltage and gate sizing under a 1.5× minimum-delay constraint, reducing energy from 72.91 to 29.165 (approximately 60% lower).
- Resulting project paper: *A New Design of 4-Bit Absolute Value Detector*, ICCSM 2025, pp. 122–127.

Millimeter-Wave Radar Based Non-Destructive Fruit Sugar Measurement — Jun 2024 – Aug 2024
Supervisor: Prof. Yong WANG, Zhejiang University
- Optimized radar data acquisition and preprocessing for non-destructive apple sweetness classification, using multi-frame sampling and temporal averaging to improve measurement stability.
- Collected training signals across apple varieties and four positions spaced 90 degrees apart on each fruit to address variety-dependent responses and within-fruit variation; contributed to a pipeline using multi-frame classification and majority voting.

Academic Projects

BERT-Based Sentiment Classification of IMDb Reviews Apr 2026 – May 2026
Instructor: Prof. Nickvash Kani

- Wrote the project code for binary sentiment classification with BERT, using the first-token representation for the classification task.
- Allocated a 512-token input budget approximately one-third to the review's beginning and two-thirds to its ending; achieved 92% classification accuracy in course-project experiments.

FPGA Game Development: *Tank Turmoil* Nov 2025 – Dec 2025
Supervisor: Prof. Zuofu Chen, UIUC

- Programmed FPGA using SystemVerilog to implement a dual-player tank battle game featuring movement, shooting, item collection, and interactive audio effects.
- Designed tank rotation, collision detection, bullet generation, VGA pixel output, and GPIO keyboard controls; simplified collision logic to reduce DSP usage while preserving intended gameplay behavior.
- Implemented a linear congruential generator on the FPGA to support randomly generated maps.

Unreal Engine 5 FPS Game Development Oct 2025 – Dec 2025
Supervisor: Prof. Eric Shaffer, UIUC

- Designed and implemented core gameplay systems, including health bars, experience and leveling mechanics, point-allocation UI, in-game shop, and purchasing interfaces.
- Developed boss AI behaviors and attack patterns, and managed multi-level scene transitions with persistent player state.
- Coordinated team activities, organized regular meetings, and tracked project progress.

Competitions

The 6th Zhejiang University Concrete Dragon Boat Competition May 2025
Supervisor: Prof. Yushi Cheng, ZJU

- Led the design of a custom concrete boat under strict dimensional constraints, balancing trade-offs among speed, maneuverability, and structural integrity.
- Awarded Third Prize at the campus level

Honors & Awards

Dean's list, UIUC, Fall 2025
Third Prize, Concrete Dragon Boat Competition, Zhejiang University May 2025
University-level Academic Excellence Award, Zhejiang University, Top 15% Dec 2024
University-level Third-class Scholarship, Zhejiang University, Top 15% Dec 2024

Technical Skills

Programming: Cuda, Python, C/C++, SystemVerilog.
Tools & Platforms:UE5, RosPy, Pytorchscikit-learn